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Project Title	**Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)**
Week	Week-2

TASK-2

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Abstract

This project presents a Dynamic Contextual Pricing Engine using the Thompson Sampling algorithm. The system simulates dynamic pricing by treating different prices as continuous bandit arms. It estimates customer purchase probabilities using contextual Beta distributions, balances exploration and exploitation, and adjusts prices dynamically in response to changing market conditions and competitive shocks.

Introduction

Dynamic pricing is widely used in online shopping, travel, and ride-sharing platforms to maximize revenue. Traditional fixed pricing cannot adapt to changing customer behavior. Thompson Sampling provides an intelligent way to learn customer preferences and choose optimal prices based on real-time conversion data while responding to market changes.

Theory

- Represents prices as continuous arms of a multi-armed bandit.
- Uses contextual Beta distributions.
- Applies Thompson Sampling.
- Balances exploration and exploitation.
- Adjusts prices according to competitive shocks.
- Improves revenue by learning from customer interactions.

Objectives

- Design a dynamic pricing simulation.
- Implement Thompson Sampling.
- Calculate posterior Beta distributions.
- Simulate customer purchasing behavior.
- Respond to competitor pricing changes.
- Maximize revenue.

Methodology

1. Initialize price range.
2. Simulate customer interactions.

3. Update Beta distributions.
4. Apply Thompson Sampling.
5. Introduce competitive shocks.
6. Evaluate revenue.

Dataset

Customer ID, Product ID, Offered Price, Purchase Decision, Conversion Rate, Customer Context, Competitor Price, Revenue.

Complexity Analysis

The pricing engine updates posterior distributions after each interaction with low computational cost, making it suitable for real-time pricing.

Results

The system adapts prices based on customer behavior and market conditions, improving expected revenue.

Conclusion

The Dynamic Contextual Pricing Engine demonstrates how Thompson Sampling optimizes pricing decisions under uncertainty.

References

- Thompson (1933).
- Reinforcement Learning: Sutton & Barto.
- Python Documentation
- NumPy Documentation
- Matplotlib Documentation